

Photosynthesis: A Short Primer

1. Overview

Photosynthesis is the process by which green plants and algae convert light energy into the chemical energy held in sugars. It draws in carbon dioxide and water, stores what it needs as glucose, and releases the rest as oxygen. Almost every food chain on the planet starts here.

2. Inputs and outputs

The reaction takes six molecules of carbon dioxide and six of water and rearranges them, with the help of sunlight caught by chlorophyll, into glucose. Six molecules of oxygen leave the leaf as a by-product.

3. Why it matters

Every gram of sugar a plant stores is energy that something else will eventually eat, which is why the productivity of a forest or an ocean is measured in carbon fixed per year.

The same reaction is what keeps atmospheric oxygen where it is, and it is the reason a warming ocean with less plankton in it is a problem far beyond the plankton.